

Lab Exercise 6

Snort IDS/IPS to detect and prevent real time threats in TryHackMe Platform

AIM:

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

SOFTWARE REQUIRED:

TryHackMe account, AttackBox or VPN, Snort, Nmap, Linux terminal.

HOW TO START TRYHACKME

1. Login to TryHackMe.
2. Search for "Snort" room.
3. Click Start Machine.
4. Click Start AttackBox.
5. Open Terminal in AttackBox.
6. Verify Snort:

```
snort -V
```

7. Test configuration:

```
sudo snort -T -c /etc/snort/snort.conf
```

Experiment 1 — ICMP (Ping) Detection (IDS) — line by line (with new terminal steps)

1. Open terminal and Check Snort:

```
snort -V
```

2. Test Configuration

```
sudo snort -T -c /etc/snort/snort.conf
```

3. Open rule File

```
sudo nano /etc/snort/rules/local.rules
```

4. Add this rule:

```
alert icmp any any -> any any (msg:"ICMP Ping Detected";  
sid:1000001; rev:1;)
```

5. Save and exit: Ctrl+O → Enter → Ctrl+X

6. Run Snort (IDS Mode)

```
sudo snort -c /etc/snort/snort.conf -i eth0
```

7. Generate Attack (New Terminal) using Open new terminal
In AttackBox:

- Click + (New Tab) at top of terminal
OR
- Right-click terminal → New Tab

8. In the new terminal:

```
ping 8.8.8.8
```

9. Go back to first terminal and View Alert → confirm alert shown
ICMP Ping Detected

10.(Log check – optional):

```
sudo tail -f /var/log/snort/alert
```

Experiment 2 — Port Scan Detection — line by line (with new terminal steps)

1. Edit Rules File

```
sudo nano /etc/snort/rules/local.rules
```

2. Add:

```
alert tcp any any -> any any (flags:S; msg:"Port Scan Detected";  
sid:1000002; rev:1;)
```

3. Save: Ctrl+O → Enter → Ctrl+X

4. Stop Snort:

```
Ctrl + C
```

5. Restart Snort:

```
sudo snort -c /etc/snort/snort.conf -i eth0
```

6. Open new terminal tab

- Click + on terminal

OR

- Right-click → New Tab

7. In new terminal: Run scan

```
nmap -sS 8.8.8.8
```

8. Go back to Snort terminal → confirm alert

9. (Optional log):

```
sudo tail -f /var/log/snort/alert
```

Experiment 3 — IPS Blocking (Inline Mode) — line by line (with new terminal steps)

1. Change rule in Rule File

```
sudo nano /etc/snort/rules/local.rules
```

2. Add:

```
drop icmp any any -> any any (msg:"ICMP Blocked"; sid:1000003;  
rev:1;)
```

3. Save: Ctrl+O → Enter → Ctrl+X

4. Stop Snort:

Ctrl + C

5. Run inline:

```
sudo snort -Q --daq afpacket -i eth0:eth0 -c /etc/snort/snort.conf
```

6. Open new terminal tab

- Click +
OR
- Right-click → New Tab

7. In new terminal:

```
ping 8.8.8.8
```

8. Expected result:

- ✗ No ping reply
- ✓ Alert shown in Snort terminal

9. (Optional log):

```
sudo tail -f /var/log/snort/alert
```

If interface is not eth0

Run once:

ip a

Use the shown interface name instead of eth0.

RESULT:

Successfully detected and prevented attacks using Snort IDS/IPS.